

Woojae Shin

KAIST

sindream.github.io
github.com/sindream
ORCID 0000-0002-6155-3712

Research interests span autonomous systems, learning-based estimation, and robotics, with public work in aerial vehicle control, radar-inertial odometry, communication relay planning, source localization, and robust regression under distribution shift.

Autonomous systems / Learning-based estimation / Aerial robotics / Sensor fusion / Robust machine learning

PUBLICATIONS

- 2026** **Vision-Based Autonomous Drone Landing on Moving Platforms With Uncertain Motion via Deep Reinforcement Learning [DOI]**
- Woojae Shin, Minwoo Kim, Taewook Park, Geunsik Bae, Seunghwan Kim, and Hyondong Oh
IEEE Robotics and Automation Letters, 11(5), 5542-5549
- 2026** **Real-Time Communication Relay Planning With a Low-Complexity Network Quality Prediction Model in Dynamic Indoor Missions [DOI]**
- Jaemin Seo, Jongyun Kim, Seunghwan Kim, Changseung Kim, **Woojae Shin**, and Hyondong Oh
IEEE Robotics and Automation Letters, 11(1), 322-329
- 2025** **EKF-Based Radar-Inertial Odometry With Online Temporal Calibration [DOI]**
- Changseung Kim, Geunsik Bae, **Woojae Shin**, Sen Wang, and Hyondong Oh
IEEE Robotics and Automation Letters, 10(7), 7230-7237
- 2024** **Autonomous Landing on a Moving Platform Using Vision-Based Deep Reinforcement Learning [DOI]**
- Pawel Ladosz, Meraj Mammadov, Heejung Shin, **Woojae Shin**, and Hyondong Oh
IEEE Robotics and Automation Letters, 9(5), 4575-4582

SELECTED PROJECTS

- STE Torch [GitHub]** Vectorized experiments for 2D gas-plume source localization, combining learned estimators, particle methods, and active mobile sensing.
Python, PyTorch, CUDA, particle filtering, active sensing
- ReDepth [GitHub]** RiTA 2024 poster code exploring rectified activations for out-of-distribution regression across synthetic, vision, and depth-estimation tasks.
Python, PyTorch, OOD robustness, depth estimation

TECHNICAL SCOPE

Python / PyTorch / CUDA / deep reinforcement learning / state estimation / active sensing / OOD robustness

Last updated October 2026. Publication metadata verified through the public ORCID and DOI records.